Ex.No: 7 Date:	TEXT – TO – SPEECH SYSTEM USING GTTS

AIM:

To create a web-based Text-to-Speech (TTS) application using gTTS and Streamlit that allow users to input text, select a language and voice speed, and convert it into audible speech.

THEORY:

- Text-to-Speech (TTS) is a technology that converts written text into spoken words using speech synthesis.
- gTTS (Google Text-to-Speech): A Python library and API interface to Google Translate's TTS engine.
- Streamlit: A Python framework to quickly build interactive web apps for data and machine learning applications.
- Users can choose different languages and voice speeds, providing flexibility in the generated audio.
- The generated audio is played directly in the web interface using Streamlit's st.audio() component.

IMPLEMENTATION:

Installation of Packages:

pip install streamlit

pip install Gtts

ALGORITHM:

Step 1: Import necessary libraries: streamlit, gTTS, tempfile, and base64.

Step 2: Set up Streamlit page with title, layout, and icon.

Step 3: Initialize session state to store user input.

Step 4: Create UI elements:

Step 5: Text area for user input.

Step 6: Sidebar to select language.

Step 7: Sidebar to select voice speed.

Step 8: Button to trigger speech conversion.

Step 9: Handle button click:

Step 10: Check if input text is empty; display a warning if yes.

Step 11: If not, generate speech using gTTS with the chosen language and speed.

Step 12: Save speech to a temporary MP3 file.

Step 13: Play audio in the app using st.audio().

Step 14: Display success message after generating audio.

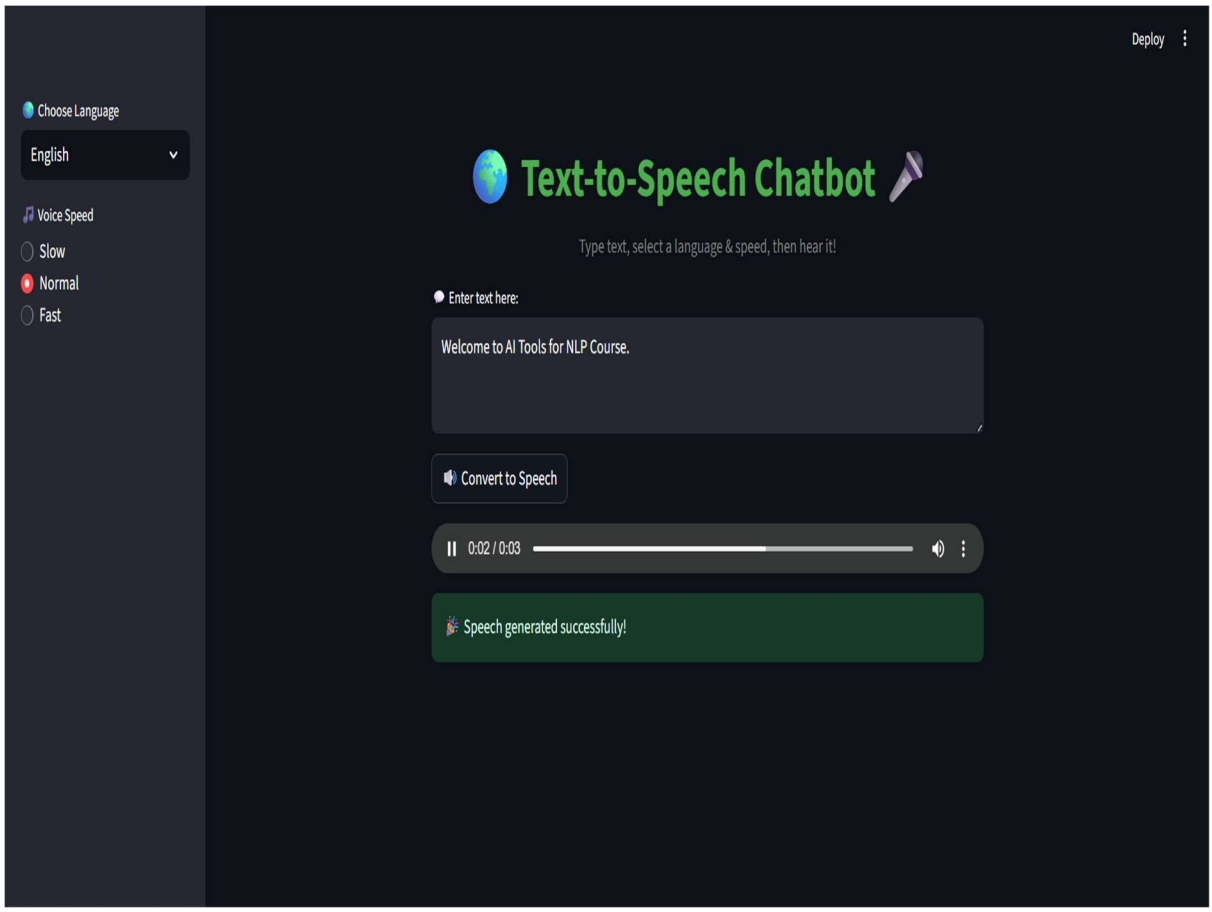
Step 15: (Optional) Clear the text input for a new entry.

CODE:

```
import streamlit as st
from gtts import gTTS
import tempfile
import base64
# Page setup
st.set_page_config(page_title="Text-to-Speech Chatbot", page_icon="🎤",
layout="centered")
# Initialize session state
if "text_input" not in st.session_state:
    st.session_state.text_input = ""
```

```
# Title
st.markdown(
    "<h1 style='text-align:center; color:#4CAF50;'>🌍 Text-to-Speech Chatbot 🎤</h1>"
    "<p style='text-align:center; color:gray;'>Type text, select a language & speed, then hear
it!</p>",
    unsafe_allow_html=True
)
# Sidebar
languages = {
    "English": "en", "French": "fr", "Spanish": "es", "German": "de",
    "Hindi": "hi", "Tamil": "ta", "Italian": "it", "Chinese": "zh-cn",
    "Japanese": "ja", "Arabic": "ar", "Russian": "ru", "Portuguese": "pt"
}
selected_lang = st.sidebar.selectbox("🌍 Choose Language", list(languages.keys()))
voice_speed = st.sidebar.radio("🎵 Voice Speed", ["Slow", "Normal", "Fast"])
# Text input
user_text = st.text_area("💬 Enter text here:", key="text_input")  # bind to session_state
# Button
if st.button("🔊 Convert to Speech"):
    if user_text.strip() == "":
        st.warning("⚠️ Please enter some text!")
    else:
        # Generate TTS
        tts = gTTS(
```

OUTPUT:


Deploy
Choose Language
English
Voice Speed
Slow
Normal
Fast
Text-to-Speech Chatbot
Type text, select a language & speed, then hear it!
Enter text here:
Welcome to AI Tools for NLP Course.
Convert to Speech
0:02 / 0:03
Speech generated successfully!

```
        text=user_text,
        lang=languages[selected_lang],
        slow=True if voice_speed == "Slow" else False
    )
    # Save temp file
    with tempfile.NamedTemporaryFile(delete=False, suffix=".mp3") as tmp_file:
        tts.save(tmp_file.name)
        audio_file = tmp_file.name
    st.audio(audio_file, format="audio/mp3")
    st.success("🎉 Speech generated successfully!")
    # Clear text input by resetting session_state variable
```

RUBRICS:

PROBLEM UNDERSTANDING AND OBJECTIVE CLARITY (20)	IMPLEMENTATION AND CODING (40)	OUTPUT EVALUATION AND DISCUSSION (30)	VIVA (10)	TOTAL (100)

RESULT:

Thus, a web-based Text-to-Speech (TTS) application using gTTS and Streamlit that allow users to input text, select a language and voice speed, and convert it into audible speech has been executed successfully.